Second, A[ ] and B[ ] must be initialized with coefficients corresponding to the *identity* system, not all zeros. This is done in lines 180 to 240. During the first loop, the coefficients for the first stage are combined with the information initially present in these arrays. If all zeros were initially present, the arrays would always remain zero. Third, two temporary arrays are used, TA[ ] and TB[ ]. These hold the old values of A[ ] and B[ ] during the convolution, freeing A[ ] and B[ ] to hold the new values.

To finish the program, block 13 (lines 540-670) adjusts the filter to have a unity gain in the passband. This operates as previously described: calculate the existing gain with Eq. 33-7 or 33-8, and divide all the "a" coefficients to normalize. The intermediate variables, SA and SB, are the sums of the "a" and "b" coefficients, respectively.

Calculate Pole Locations in the s-Plane

Regardless of the type of filter being designed, this program begins with a Butterworth low-pass filter in the *s-plane*, with a cutoff frequency of $\omega = 1$. As described in the last chapter, Butterworth filters have poles that are equally spaced around a circle in the s-plane. Since the filter is low-pass, no zeros are used. The radius of the circle is *one*, corresponding to the cutoff frequency of $\omega = 1$. Block 3 of the flowchart (lines 1080 & 1090) calculate the location of each pole-pair in rectangular coordinates. The program variables, RP and IP, are the real and imaginary parts of the pole location, respectively. These program variables correspond to σ and ω, where the pole-pair is located at $\sigma \pm j\omega$. This pole location is calculated from the number of poles in the filter and the stage being worked on, the program variables: NP and P%, respectively.

Warp from Circle to Ellipse

To implement a Chebyshev filter, this *circular* pattern of poles must be transformed into an *elliptical* pattern. The relative flatness of the ellipse determines how much ripple will be present in the passband of the filter. If the pole location on the circle is given by: σ and ω, the corresponding location on the ellipse, σ' and ω', is given by:

$$\sigma' = \sigma \sinh(v)/k$$

$$\omega' = \omega \cosh(v)/k$$

where:

$$v = \frac{\sinh^{-1}(1/\epsilon)}{NP}$$

$$k = \cosh\left(\frac{1}{NP}\cosh^{-1}\frac{1}{\epsilon}\right)$$

$$\epsilon = \left[\left(\frac{100}{100 - PR}\right)^2 - 1\right]^{1/2}$$

EQUATION 33-9
Circular to elliptical transform. These equations change the pole location on a circle to a corresponding location on an ellipse. The variables, *NP* and *PR*, are the number of poles in the filter, and the percent ripple in the passband, respectively. The location on the circle is given by σ and ω, and the location on the ellipse by σ' and ω'. The variables ϵ, v, and k, are used only to make the equations shorter.